

EX No:1	EDA-DATA CLEANING

Aim:

To perform Exploratory Data Analysis (EDA) and data cleaning on the tourism dataset using standard data science frameworks (CRISP-DM, SEMMA, and KDD) by handling missing values, removing duplicates, creating derived features, and analyzing tourism patterns.

Algorithm:

1. Import pandas, numpy, and matplotlib.pyplot.
2. Load tourism_dataset_200.csv and inspect dataset structure, missing values, and duplicates.
3. Clean data by removing duplicates and handling missing values in numerical columns.
4. Create a derived feature Budget_Per_Day_USD using estimated budget and average duration.
5. Plot frequency distributions for key tourism variables and display a correlation matrix heatmap.
6. Summarize tourism insights and end execution.

Program:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

# CRISP-DM FRAMEWORK

# =====

# 1. BUSINESS UNDERSTANDING

# Objective:

# Analyze tourism destinations, travel preferences,

# budgets, ratings, and visitor patterns.

# 2. DATA UNDERSTANDING

df = pd.read_csv("tourism_dataset_200.csv")

print(df.head())

print(df.shape)

print(df.info())

print(df.describe())

print(df.isnull().sum())

print(df.duplicated().sum())

# 3. DATA PREPARATION

df.drop_duplicates(inplace=True)

df['Rating'] = df['Rating'].fillna(df['Rating'].median())

df['Estimated_Budget_USD'] = df['Estimated_Budget_USD'].fillna(

df['Estimated_Budget_USD'].median()

)

df['Annual_Visitors'] = df['Annual_Visitors'].fillna(

df['Annual_Visitors'].median()

)

# Derived feature

df['Budget_Per_Day_USD'] = (

df['Estimated_Budget_USD'] / df['Average_Duration_Days']

).round(2)

# 4. MODELING (EDA)

df['Tourism_Type'].value_counts().plot(kind='bar')
```

```

plt.title("Tourism Type Distribution")

plt.xlabel("Tourism Type")

plt.ylabel("Count")

plt.show()

df['Continent'].value_counts().plot(kind='bar')

plt.title("Tourists Destinations by Continent")

plt.xlabel("Continent")

plt.ylabel("Count")

plt.show()

df['Best_Season'].value_counts().plot(kind='bar')

plt.title("Best Season for Tourism")

plt.xlabel("Season")

plt.ylabel("Count")

plt.show()

df.groupby('Country')['Estimated_Budget_USD'].mean().sort_values(
ascending=False
).head(10).plot(kind='bar')

plt.title("Top 10 Countries by Average Estimated Budget")

plt.xlabel("Country")

plt.ylabel("Average Budget (USD)")

plt.show()

corr = df.select_dtypes(include='number').corr()

plt.figure(figsize=(10, 8))

plt.imshow(corr)

plt.colorbar()

plt.xticks(range(len(corr.columns)), corr.columns, rotation=90)

plt.yticks(range(len(corr.columns)), corr.columns)

plt.title("Correlation Heatmap")

plt.show()

# 5. EVALUATION

print("Tourism insights generated successfully.")

# 6. DEPLOYMENT

print("Insights can be deployed through tourism dashboards.")

```

```

# SEMMA FRAMEWORK

# =====

# SAMPLE

df = pd.read_csv("tourism_dataset_200.csv")

# EXPLORE

print(df.head())

print(df.info())

print(df.describe())

print(df.isnull().sum())

# MODIFY

df.drop_duplicates(inplace=True)

df['Rating'] = df['Rating'].fillna(df['Rating'].median())

df['Estimated_Budget_USD'] = df['Estimated_Budget_USD'].fillna(
df['Estimated_Budget_USD'].median()
)

df['Annual_Visitors'] = df['Annual_Visitors'].fillna(
df['Annual_Visitors'].median()
)

df['Budget_Per_Day_USD'] = (
df['Estimated_Budget_USD'] / df['Average_Duration_Days']
).round(2)

# MODEL

df['Tourism_Type'].value_counts().plot(kind='bar')

plt.show()

df['Continent'].value_counts().plot(kind='bar')

plt.show()

df['Best_Season'].value_counts().plot(kind='bar')

plt.show()

df.groupby('Country')['Estimated_Budget_USD'].mean().sort_values(
ascending=False
).head(10).plot(kind='bar')

plt.show()

corr = df.select_dtypes(include='number').corr()

```

```

plt.figure(figsize=(10, 8))

plt.imshow(corr)

plt.colorbar()

plt.xticks(range(len(corr.columns)), corr.columns, rotation=90)

plt.yticks(range(len(corr.columns)), corr.columns)

plt.show()

# ASSESS

print("Tourism patterns and trends successfully identified.")

# KDD FRAMEWORK

# =====

# SELECTION

df = pd.read_csv("tourism_dataset_200.csv")

# PREPROCESSING

print(df.info())

print(df.isnull().sum())

print(df.duplicated().sum())

df.drop_duplicates(inplace=True)

df['Rating'] = df['Rating'].fillna(df['Rating'].median())

df['Estimated_Budget_USD'] = df['Estimated_Budget_USD'].fillna(
df['Estimated_Budget_USD'].median()
)

df['Annual_Visitors'] = df['Annual_Visitors'].fillna(
df['Annual_Visitors'].median()
)

# TRANSFORMATION

df['Budget_Per_Day_USD'] = (
df['Estimated_Budget_USD'] / df['Average_Duration_Days']
).round(2)

# DATA MINING

df['Tourism_Type'].value_counts().plot(kind='bar')

plt.show()

df['Continent'].value_counts().plot(kind='bar')

plt.show()

```

```
df['Best_Season'].value_counts().plot(kind='bar')

plt.show()

df.groupby('Country')['Estimated_Budget_USD'].mean().sort_values(
ascending=False
).head(10).plot(kind='bar')

plt.show()

corr = df.select_dtypes(include='number').corr()

plt.figure(figsize=(10, 8))

plt.imshow(corr)

plt.colorbar()

plt.xticks(range(len(corr.columns)), corr.columns, rotation=90)

plt.yticks(range(len(corr.columns)), corr.columns)

plt.show()

# INTERPRETATION

print("Knowledge discovered from the tourism dataset has been interpreted to
support tourism planning and business decisions.")
```

Output:

	Tourism_ID	Destination	Country	Continent	Tourism_Type	Best_Season	\
0	T001	Los Angeles	USA	North America	City	Winter	
1	T002	Barcelona	Spain	Europe	City	Winter	
2	T003	Paris	France	Europe	Wildlife	Winter	
3	T004	Singapore	Singapore	Asia	Wildlife	Spring	
4	T005	Tokyo	Japan	Asia	Cultural	Spring	

	Average_Duration_Days	Estimated_Budget_USD	Rating	Annual_Visitors	\
0	3	504	4.6	133393	
1	11	3756	3.5	54123	
2	10	1928	4.6	372696	
3	14	353	4.6	88707	
4	14	3057	3.7	204191	

	Accommodation_Type	Transportation
0	Hotel	Cruise
1	Apartment	Cruise
2	Homestay	Cruise
3	Hostel	Train
4	Hostel	Cruise

** RangeIndex: 200 entries, 0 to 199

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	Tourism_ID	200 non-null	object
1	Destination	200 non-null	object
2	Country	200 non-null	object
3	Continent	200 non-null	object
4	Tourism_Type	200 non-null	object
5	Best_Season	200 non-null	object
6	Average_Duration_Days	200 non-null	int64
7	Estimated_Budget_USD	200 non-null	int64
8	Rating	200 non-null	float64
9	Annual_Visitors	200 non-null	int64
10	Accommodation_Type	200 non-null	object
11	Transportation	200 non-null	object

dtypes: float64(1), int64(3), object(8)

memory usage: 18.9+ KB

None

	Average_Duration_Days	Estimated_Budget_USD	Rating	Annual_Visitors
count	200.000000	200.000000	200.000000	200.000000
mean	8.185000	2841.885000	4.26450	254984.850000
std	3.916307	1351.514379	0.44018	137082.937526
min	2.000000	345.000000	3.50000	5212.000000
25%	5.000000	1725.000000	3.90000	145942.500000
50%	8.000000	2862.000000	4.20000	253579.000000
75%	12.000000	4036.750000	4.60000	363804.250000
max	14.000000	4962.000000	5.00000	499268.000000

company	82137
agent	12193
country	452
children	4
reserved_room_type	0
assigned_room_type	0
booking_changes	0
deposit_type	0
hotel	0
previous_cancellations	0
days_in_waiting_list	0
customer_type	0
adr	0
required_car_parking_spaces	0
total_of_special_requests	0
reservation_status	0
previous_bookings_not_canceled	0
is_repeated_guest	0
is_canceled	0
distribution_channel	0
market_segment	0
meal	0
babies	0
adults	0
stays_in_week_nights	0
stays_in_weekend_nights	0
arrival_date_day_of_month	0
arrival_date_week_number	0
arrival_date_month	0
arrival_date_year	0
lead_time	0
reservation_status_date	0

dtype: int64

◀

▶

Tourism patterns and trends successfully identified.

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 200 entries, 0 to 199

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	Tourism_ID	200 non-null	object
1	Destination	200 non-null	object
2	Country	200 non-null	object
3	Continent	200 non-null	object
4	Tourism_Type	200 non-null	object
5	Best_Season	200 non-null	object
6	Average_Duration_Days	200 non-null	int64
7	Estimated_Budget_USD	200 non-null	int64
8	Rating	200 non-null	float64
9	Annual_Visitors	200 non-null	int64
10	Accommodation_Type	200 non-null	object
11	Transportation	200 non-null	object

dtypes: float64(1), int64(3), object(8)

memory usage: 18.9+ KB

Result:

Exploratory Data Analysis and data cleaning were successfully performed on the tourism dataset using CRISP-DM, SEMMA, and KDD frameworks. The dataset was inspected, cleaned, transformed, and analyzed to identify patterns related to tourism type, continent, season, budget, duration, ratings, and annual visitors.